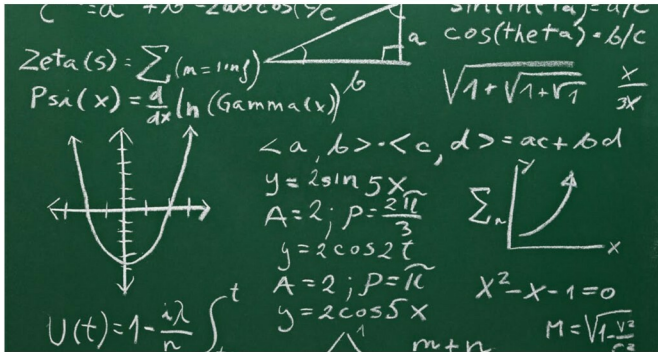


Lists: The Building Blocks of Maxima

This 20th article in our series on Mathematics in Open Source showcases the list manipulations in Maxima, the programming language with an ALGOL-like syntax but Lisp-like semantics.



Lists are the basic building blocks of Maxima. The fundamental reason is that Maxima is implemented in Lisp, the building blocks of which are also lists.

To begin with, let us walk through the ways of creating a list. The simplest method to get a list in Maxima is to just define it, using []. So, $[x, 5, 3, 2*y]$ is a list consisting of four members. However, Maxima provides two powerful functions for automatically generating lists: *makelist()* and *create_list()*.

makelist() can take two forms. *makelist(e, x, x0, xn)* creates and returns a list using the expression 'e', evaluated for 'x' using the values ranging from 'x0' to 'xn'. *makelist(e, x, L)* creates and returns a list using the expression 'e', evaluated for 'x' using the members of the list L. Check out the example below for better clarity:

```
$ maxima -q
(%i1) makelist(2 * i, i, 1, 5);
(%o1) [2, 4, 6, 8, 10]
(%i2) makelist(concat(x, 2 * i - 1), i, 1, 5);
(%o2) [x1, x3, x5, x7, x9]
(%i3) makelist(concat(x, 2), x, [a, b, c, d]);
(%o3) [a2, b2, c2, d2]
(%i4) quit();
```

Note the interesting usage of *concat()* to just concatenate

its arguments. Note that *makelist()* is limited by the variation it can have, which to be specific, is just one – 'i' in the first two examples and 'x' in the last one. If we want more, the *create_list()* function comes into play.

create_list(f, x1, L1, ..., xn, Ln) creates and returns a list with members of the form 'f', evaluated for the variables x1, ..., xn using the values from the corresponding lists L1, ..., Ln. Here is just a glimpse of its power:

```
$ maxima -q
(%i1) create_list(concat(x, y), x, [p, q], y, [1, 2]);
(%o1) [p1, p2, q1, q2]
(%i2) create_list(concat(x, y, z), x, [p, q], y, [1, 2], z, [a, b]);
(%o2) [p1a, p1b, p2a, p2b, q1a, q1b, q2a, q2b]
(%i3) create_list(concat(x, y, z), x, [p, q], y, [1, 2, 3], z, [a, b]);
(%o3) [p1a, p1b, p2a, p2b, p3a, p3b, q1a, q1b, q2a, q2b, q3a, q3b]
(%i4) quit();
```

Note that 'all possible combinations' are created using the values for the variables 'x', 'y' and 'z'.

Once we have created the lists, Maxima provides a host of functions to play around with them. Let's take a look at these.